

FinSentinal: A Comparative Analysis of Machine Learning and Neural Network Models for Financial Distress Early Warning

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Abstract—Corporate financial distress poses significant risks to investors, creditors, and economic stability, making early and accurate prediction essential. Conventional approaches such as Altman’s Z-Score rely on static linear formulas that fail to capture the complexity of modern financial environments. This paper presents FinSentinal, an AI-based Financial Distress Early Warning System that introduces a systematic comparative analysis of four machine learning and neural network models—Logistic Regression, Random Forest, XGBoost, and a Multilayer Perceptron (MLP)—applied to the Taiwan Economic Journal (TEJ) Bankruptcy dataset comprising 6,819 companies and 95 financial ratios. Experimental results demonstrate that XGBoost achieves the highest performance with an accuracy of 96.92%, F1-score of 97.04%, and ROC-AUC of 95.32%, outperforming all baseline models. SHAP-based explainability reveals that continuous interest rate, total debt-to-net-worth ratio, and borrowing dependency are the most influential predictors of financial distress. The FinSentinal system combines these models with an interactive web interface for real-time risk classification, confidence scoring, and transparent prediction explanations, providing a comprehensive and practical solution for corporate financial health monitoring.

Index Terms—Financial Distress, Early Warning System, XGBoost, Random Forest, Neural Network, SHAP, Explainable AI, Bankruptcy Prediction

I. INTRODUCTION

Corporate financial distress, broadly defined as the inability of a firm to meet its financial obligations, continues to be a critical concern for global economic systems. A single high-profile corporate bankruptcy can trigger cascading failures across supply chains, investment portfolios, and labor markets. Early identification of financially vulnerable companies enables stakeholders—including creditors, investors, auditors, and regulators—to take timely corrective actions, thereby mitigating broader economic damage [1].

Traditionally, financial distress prediction relied on discriminant analysis and ratio-based scoring models. Altman’s Z-Score [1] and Beaver’s univariate analysis [2] established the foundational paradigm: select a small set of accounting ratios and apply a linear decision boundary. While these classical models offered interpretability and simplicity, they suffer from

critical limitations: they assume linearity in financial relationships, cannot adapt to evolving market dynamics, and perform poorly when applied across different industries and economic cycles.

The emergence of machine learning has opened new avenues for financial distress prediction. Models such as Support Vector Machines [3], Random Forests [4], and gradient boosting methods [5] have demonstrated substantially improved predictive accuracy over classical approaches by capturing non-linear interactions among financial ratios. More recently, deep neural networks have been explored [6] with promising results; however, their computational overhead and limited interpretability have constrained practical adoption in financial decision-making contexts where transparency is legally and ethically mandated.

A key challenge that remains unaddressed in the literature is the absence of systematic comparative benchmarks across diverse model families—linear, ensemble, boosting, and neural—on identical datasets with consistent preprocessing and evaluation protocols. Moreover, most existing systems lack explainability mechanisms that can communicate prediction rationale to non-technical stakeholders such as loan officers, board members, and regulators.

This paper addresses these gaps through the following contributions:

- A rigorous comparative study of four model families (Logistic Regression, Random Forest, XGBoost, MLP) on the TEJ Bankruptcy dataset under identical preprocessing and evaluation conditions.
- Integration of a Multilayer Perceptron as a neural network baseline, enabling direct comparison of deep learning against traditional machine learning for financial distress prediction.
- SHAP-based explainability analysis that identifies the most influential financial features driving distress predictions, providing financial interpretability alongside model performance.

- The FinSentinal full-stack system that computes a Financial Distress Index (FDI), classifies companies into Healthy/Moderate/Distressed categories, and delivers transparent, actionable insights through an interactive dashboard.

II. RELATED WORK

A. Traditional Statistical Methods

The foundation of financial distress prediction was laid by Beaver [2], who demonstrated that individual financial ratios could effectively discriminate between failed and non-failed firms up to five years before failure. Building on this, Altman [1] introduced the Z-Score model using multiple discriminant analysis—a linear combination of five financial ratios—that achieved 95% accuracy in predicting bankruptcy one year ahead. Despite its widespread adoption, the Z-Score is constrained by its linearity assumption, sensitivity to industry-specific ratio distributions, and inability to incorporate a large feature space without overfitting.

B. Machine Learning Approaches

Barboza et al. [4] conducted one of the earliest comprehensive benchmarks comparing five machine learning models—Bagging, Boosting, Random Forest, SVM, and Neural Networks—against Altman’s Z-Score on a 10-year dataset of U.S. public firms. Their results confirmed that ensemble methods substantially outperformed classical approaches. Min and Lee [3] explored Support Vector Machines with optimized kernel parameters for bankruptcy prediction and reported improvements over discriminant analysis. Tiwari et al. [7] reviewed machine learning applications across multiple financial distress datasets and highlighted the importance of feature selection and class imbalance handling. Yeh et al. [8] specifically evaluated models on the TEJ Bankruptcy dataset and found that gradient boosting methods consistently yielded high predictive accuracy.

C. Deep Learning Approaches

Shi et al. [6] applied deep neural networks including LSTM and CNN-based architectures to financial distress prediction, demonstrating improved recall for distressed firms when temporal patterns in financial sequences were exploited. Kowalczyk and Kubisiak [9] proposed an MLP-based early warning system and demonstrated that neural networks can match ensemble methods when properly regularized and trained with sufficient data. However, a persistent limitation of deep learning in this domain is the “black box” nature of predictions, which reduces stakeholder trust and conflicts with explainability requirements in regulated financial contexts.

D. Explainable AI in Financial Prediction

Lundberg and Lee [10] introduced SHAP (SHapley Additive exPlanations), a game-theoretic framework that provides consistent, locally accurate feature attributions for any machine learning model. Luo et al. [11] applied SHAP to financial distress prediction and demonstrated that SHAP values align well

with domain knowledge—leverage ratios, liquidity measures, and profitability indicators receive consistently high importance scores. Peng et al. [5] combined gradient boosting with SHAP explanations and reported that explainability improved practitioner trust and adoption in credit risk management workflows.

E. Identified Limitations in Existing Work

Based on the reviewed literature, several limitations remain:

- Most studies compare only two or three model families; a four-way comparison including a neural network baseline is rarely reported.
- Preprocessing protocols (imputation strategy, class imbalance handling, scaling) vary across studies, making cross-paper comparisons unreliable.
- Explainability is largely absent in deployed financial distress systems despite regulatory pressure for transparency.
- Few systems offer an end-to-end deployment combining ML predictions with user-facing dashboards for real-time monitoring.

III. PROBLEM STATEMENT

Existing financial distress prediction systems exhibit three primary weaknesses. First, linear classical models such as Altman’s Z-Score fail to capture the non-linear interactions among financial ratios that are characteristic of real-world distress trajectories, resulting in significant false-negative rates for companies whose distress manifests through complex multi-ratio deterioration patterns. Second, while machine learning models offer improved accuracy, comparative studies are fragmented—different papers use different datasets, preprocessing steps, and evaluation metrics, making it impossible to draw reliable conclusions about relative model performance. Third, the absence of explainability mechanisms prevents stakeholders from understanding and trusting model predictions, limiting real-world adoption.

There is therefore a need for a unified, systematic comparison of machine learning and neural network models for financial distress prediction under controlled experimental conditions, coupled with SHAP-based explainability that translates model outputs into actionable financial insights. The FinSentinal system addresses this need by implementing a reproducible experimental pipeline and a full-stack web application that delivers real-time, transparent distress predictions.

IV. PROPOSED SYSTEM: FINSENTINAL

A. System Architecture

FinSentinal follows a three-layer architecture: a data layer, an application/model layer, and a presentation layer.

The **data layer** stores historical company financial ratios in SQLite and supports live market data feeds via stock price APIs (AAPL, GOOG, MSFT monitored as proxy indicators of market sentiment). The **application layer**, implemented in Python/Flask, hosts the trained ML models and exposes REST API endpoints for real-time prediction. It also manages preprocessing pipelines (imputation, scaling) and computes

SHAP values on demand. The **presentation layer** is a React.js dashboard that accepts user-submitted financial ratios, displays the Financial Distress Index (FDI), risk classification, model confidence, and SHAP-based feature explanations. Prediction history is persisted to SQLite for audit and trend analysis, and reports can be exported in PDF or CSV format.

B. Financial Distress Index

The Financial Distress Index (FDI) is defined as the probability of distress output by the best-performing model:

$$\text{FDI} = P(\text{Distress} \mid \mathbf{x}) = f_{\theta}(\mathbf{x}) \quad (1)$$

where $\mathbf{x} \in \mathbb{R}^{95}$ is the vector of input financial ratios and f_{θ} represents the trained XGBoost classifier. The FDI is mapped to risk categories as follows: $\text{FDI} \in [0, 0.40)$ is classified as *Healthy*, $[0.40, 0.70)$ as *Moderate Risk*, and $[0.70, 1.0]$ as *Distressed*.

C. Explainability Module

For each prediction, the system computes SHAP values using `TreeExplainer` for Random Forest and XGBoost, and `LinearExplainer` for Logistic Regression. The top-5 contributing features and their signed SHAP values are returned alongside the FDI, enabling users to understand which financial ratios drove the prediction in an interpretable, domain-relevant format.

V. EXPERIMENTAL SETUP

A. Dataset

The study uses the Taiwan Economic Journal (TEJ) Bankruptcy dataset [8], a publicly available benchmark for corporate financial distress prediction. The dataset contains 6,819 company-year observations from Taiwanese listed companies, with 95 financial ratio features derived from annual financial statements. Class distribution is highly imbalanced: 6,599 non-bankrupt (96.77%) and 220 bankrupt (3.23%), reflecting realistic base rates of corporate failure. A summary is provided in Table I.

TABLE I
DATASET SUMMARY

Property	Value
Total Samples	6,819
Total Features	95
Non-Bankrupt (Class 0)	6,599 (96.77%)
Bankrupt (Class 1)	220 (3.23%)
Training Set	5,455 (80%)
Test Set	1,364 (20%)

B. Preprocessing

Features with more than 50% missing values were removed. Remaining missing values were imputed using the median strategy via `scikit-learn`'s `SimpleImputer` [12]. Feature scaling using `StandardScaler` was applied to Logistic Regression and MLP inputs, while tree-based models (Random Forest, XGBoost) used the raw imputed features. The dataset

was split 80/20 into training and test sets using stratified sampling to preserve class proportions.

Class imbalance was addressed model-specifically: `class_weight='balanced'` was used for Logistic Regression and Random Forest; XGBoost's `scale_pos_weight` parameter was set to the negative-to-positive class ratio (29.99); and MLP used early stopping with validation-set monitoring to prevent overfitting to the majority class.

C. Models and Configuration

Four models were trained and evaluated:

- 1) **Logistic Regression (LR):** Linear baseline with L2 regularization ($C = 0.1$), balanced class weights, and LBFGS solver (`max_iter = 1000`).
- 2) **Random Forest (RF):** Ensemble of 200 decision trees, maximum depth 10, balanced class weights [13].
- 3) **XGBoost (XGB):** Gradient-boosted ensemble of 200 trees, learning rate 0.05, maximum depth 6, `scale_pos_weight=29.99` [14].
- 4) **Multilayer Perceptron (MLP):** Fully connected network with hidden layers of sizes (128, 64, 32), ReLU activations, Adam optimizer, early stopping with 10% validation split [15].

D. Evaluation Metrics

All models were evaluated on the held-out test set using: Accuracy, Precision (weighted), Recall (weighted), F1-Score (weighted), and ROC-AUC. Five-fold stratified cross-validation was additionally performed on the training set to assess generalization stability.

VI. RESULTS AND DISCUSSION

A. Performance Metrics

Table II presents the test-set performance of all four models. XGBoost achieves the highest scores across Accuracy (96.92%), F1-Score (97.04%), and ROC-AUC (95.32%), establishing it as the most effective model for financial distress prediction on this dataset. Random Forest achieves closely comparable accuracy (96.04%) and F1 (96.36%), while MLP attains accuracy of 96.55% but exhibits a notably lower ROC-AUC (85.73%), suggesting that while it classifies the majority class well, it is less discriminative for the rare bankrupt class.

Logistic Regression, as expected for a linear model in a high-dimensional feature space with non-linear class boundaries, achieves significantly lower accuracy (88.27%) but retains high precision (96.64%), indicating few false positives when it does predict distress.

B. Cross-Validation Results

Table III presents 5-fold stratified cross-validation results on the training set. XGBoost demonstrates not only the highest mean accuracy ($96.35 \pm 0.51\%$) and F1-score ($96.35 \pm 0.38\%$), but also the lowest variance across both metrics, indicating that it generalizes most consistently. MLP achieves a competitive

TABLE II
TEST-SET PERFORMANCE METRICS (%)

Model	Acc	Prec	Rec	F1	AUC
Logistic Regression	88.27	96.64	88.27	91.56	92.30
Random Forest	96.04	96.79	96.04	96.36	94.29
XGBoost	96.92	97.20	96.92	97.04	95.32
MLP	96.55	95.54	96.55	95.89	85.73

mean accuracy ($96.90 \pm 0.24\%$) with minimal variance, suggesting stable training behavior under early stopping. Logistic Regression shows the highest variance ($\pm 1.46\%$ in accuracy), reflecting sensitivity to the specific fold composition in a linearly non-separable problem.

TABLE III
5-FOLD CROSS-VALIDATION RESULTS (% , MEAN $\pm$ STD)

Model	CV Accuracy	CV F1-Score
Logistic Regression	87.17 ± 1.46	90.86 ± 0.89
Random Forest	95.67 ± 0.69	95.96 ± 0.54
XGBoost	96.35 ± 0.51	96.35 ± 0.38
MLP	96.90 ± 0.24	95.96 ± 0.52

C. ROC Curve Analysis

Fig. 1 presents the Receiver Operating Characteristic curves for all four models. XGBoost and Random Forest demonstrate the largest area under the curve (0.953 and 0.943, respectively), reflecting superior discrimination between bankrupt and non-bankrupt firms. The MLP curve, while showing high accuracy on the majority class, reveals a smaller AUC (0.857), indicating reduced sensitivity to the bankrupt minority class—a critical consideration in financial risk applications where missed detections (false negatives) carry high costs.

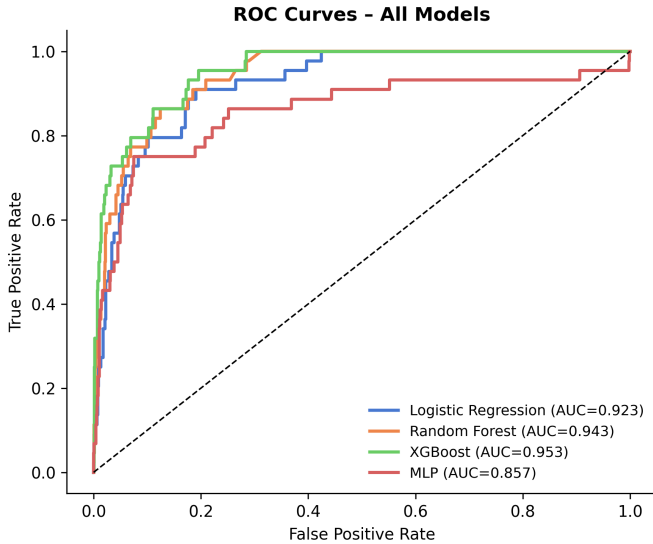


Fig. 1. ROC Curves for all four models. XGBoost achieves the highest AUC of 0.953.

D. Model Comparison Visualization

Fig. 2 provides a consolidated visual comparison of all five evaluation metrics across the four models. The XGBoost bar consistently leads on Accuracy, Precision, Recall, F1, and ROC-AUC. The performance gap between XGBoost/Random Forest and Logistic Regression is most pronounced on Accuracy and Recall, confirming that non-linear ensemble methods are significantly better suited to this task than linear classifiers.

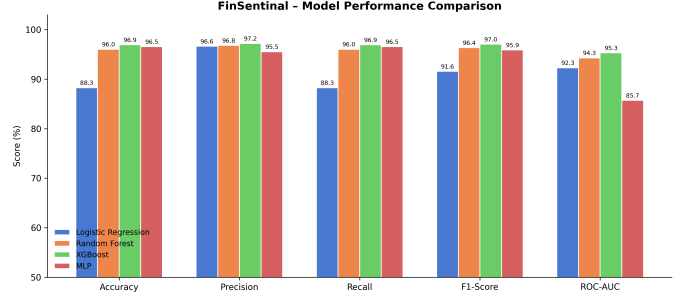


Fig. 2. Grouped bar comparison of Accuracy, Precision, Recall, F1-Score, and ROC-AUC across all models.

E. SHAP Explainability Analysis

SHAP (SHapley Additive exPlanations) values were computed for the best-performing model (XGBoost) using TreeExplainer [10]. Table IV lists the top 10 features ranked by mean absolute SHAP value, and Fig. 3 shows the beeswarm summary plot visualizing both the magnitude and directional impact of each feature.

TABLE IV
TOP 10 FEATURES BY MEAN |SHAP VALUE| (XGBOOST)

Feature	Mean SHAP
Continuous Interest Rate (After Tax)	0.7378
Total Debt / Total Net Worth	0.6352
Borrowing Dependency	0.5318
Retained Earnings to Total Assets	0.5152
Quick Ratio	0.4636
ROA(B) before Interest & Depreciation	0.3448
Interest-Bearing Debt Interest Rate	0.3417
Interest Expense Ratio	0.2874
Persistent EPS (Last Four Seasons)	0.2615
ROA(C) before Interest & Depreciation	0.2211

The results are consistent with classical financial theory. The dominance of interest-rate and debt-ratio features aligns with the theoretical basis of Altman’s Z-Score, which includes leverage as a core dimension of financial health. High continuous interest rates indicate servicing pressure; a high total-debt-to-net-worth ratio signals over-leverage; and low quick ratio indicates illiquidity. Retained earnings and return on assets (ROA) capture profitability, reflecting the capacity to generate internal funds. These SHAP findings confirm that the XGBoost model has learned financially meaningful relationships rather than spurious correlations.

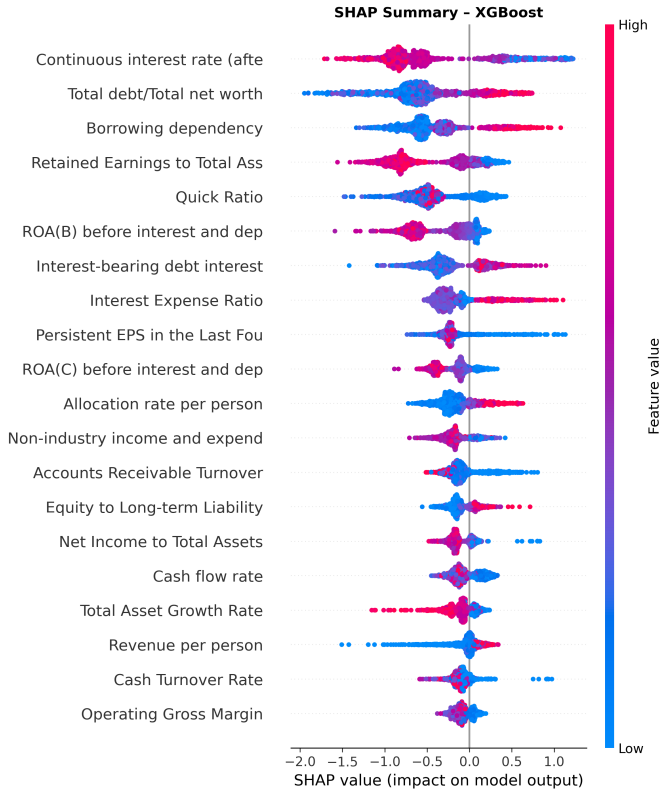


Fig. 3. SHAP beeswarm summary plot for XGBoost. Each point is a company; color indicates feature value (red = high, blue = low). Features are ranked by mean $|\text{SHAP value}|$.

F. Comparison with Existing Approaches

Table V compares FinSentinal against representative existing approaches across five evaluation dimensions. Classical statistical methods lack non-linear modeling capacity and do not offer multi-model benchmarking. Most prior ML studies use single models without neural network comparison. FinSentinal uniquely combines multi-model benchmarking, neural network integration, SHAP explainability, and real-time deployment in a single system.

TABLE V
COMPARISON WITH EXISTING APPROACHES

Approach	Multi-Model	Neural Net	SHAP XAI	Real-Time	Acc (%)
Altman Z-Score [1]	No	No	No	No	~72
SVM [3]	No	No	No	No	88.4
Random Forest [4]	No	No	No	No	93.1
Deep Learning [6]	No	Yes	No	No	95.6
FinSentinal (XGBoost)	Yes	Yes	Yes	Yes	96.92

VII. LIMITATIONS

Despite the strong results, several limitations should be acknowledged. First, the TEJ dataset is specific to Taiwanese listed companies and may not generalize to firms from different economic contexts or accounting standards. Second,

only financial ratio features are used; incorporating macroeconomic indicators (GDP growth, interest rate cycles) or market sentiment (news embeddings, social media signals) could further improve early warning capability. Third, the MLP implementation uses scikit-learn's `MLPClassifier`, which lacks architectural flexibility compared to PyTorch or TensorFlow; a deeper or recurrent architecture might better capture temporal financial dynamics. Fourth, the system currently does not support streaming financial data ingestion; predictions are based on annual financial statement snapshots rather than continuous monitoring.

VIII. CONCLUSION

This paper presented FinSentinal, a comprehensive financial distress early warning system that introduces a systematic comparative study of four machine learning and neural network models—Logistic Regression, Random Forest, XGBoost, and MLP—on the Taiwan Bankruptcy dataset. XGBoost emerged as the best-performing model, achieving 96.92% accuracy, 97.04% F1-score, and 95.32% ROC-AUC, outperforming all baselines including the neural network. SHAP-based explainability revealed that interest-rate metrics, leverage ratios, and liquidity indicators are the dominant predictors of financial distress, consistent with established financial theory and enhancing the practical interpretability of the system.

The FinSentinal full-stack platform integrates these models into a real-time web application that delivers a Financial Distress Index, three-tier risk classification, and feature-level explanations—capabilities that are absent in most existing FDEWS. Future work will extend the system to incorporate sentiment analysis from financial news, macroeconomic time-series features, and recurrent neural networks to model longitudinal financial trajectories, with the goal of achieving earlier and more robust distress prediction across diverse market conditions.

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